

The KT-D1V5 Finite-Volume Discrete Boltzmann Solver

Method, Verification on the Sod Shock Tube, and Comparison with
Conventional Shock-Capturing Schemes

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Abstract. This report describes a one-dimensional finite-volume discrete Boltzmann method (FVDBM) for compressible flow and evaluates it on the Sod shock tube. The gas is represented by five discrete particle populations (the Kataoka–Tsutahara D1V5 model) that are transported through a finite-volume grid and relaxed toward a local equilibrium by a BGK collision term. Spatial accuracy is provided by MUSCL reconstruction with a generalized-minmod limiter ($\theta = 1.2$), and time integration by a three-stage strong-stability-preserving Runge–Kutta scheme. On five grids from 3,125 to 50,000 cells the solution converges without oscillation, conserves mass and energy to 10^{-12} , and shows observed orders of 0.81–0.89 in the L_1 norm and 0.40–0.63 in the L_2 norm. These rates are below the smooth-flow values of 1 (first order) and 2 (second order), as the theory for solutions containing a shock and a contact discontinuity predicts for every shock-capturing method. The solver is compared, on identical grids, with first-order and other limited versions of itself and with a conventional finite-volume Euler solver. At equal resolution the DBM and a conventional second-order Euler solver reach similar accuracy and convergence rates, clearly better than first-order methods, but the DBM requires 100 to 800 times more computing time because its explicit collision term limits the time step to one quarter of the relaxation time.

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1 Introduction

Discrete Boltzmann methods (DBMs) model a gas with a small number of particle populations rather than solving the fluid equations directly. The macroscopic density, velocity, temperature and pressure are obtained as sums (moments) of the populations, and the familiar Euler equations are recovered when collisions are frequent [9, 21]. The approach is attractive because the same populations also carry information about departures from thermodynamic equilibrium, which conventional Euler or Navier–Stokes solvers do not represent.

The solver examined here solves the one-dimensional Sod shock tube [15], a standard benchmark whose exact solution contains the three basic wave types of gas dynamics. The objectives of this report are to

1. describe the physical model and the numerical method in a concise, self-contained way;
2. quantify the accuracy of the solver and state clearly what order of accuracy the measurements indicate;
3. compare its accuracy and computational cost with alternative methods run on the same problem and grids; and
4. explain each figure and what it implies for the method.

All figures were produced in MATLAB and are stored, in `.fig`, `.png` and `.pdf` form, in the `figures` folder of the research directory.

Configuration. Unless stated otherwise, all results use the settings in Table 1.

Table 1. Solver configuration.

Parameter	Symbol	Value
Ratio of specific heats	γ	1.4
Relaxation time	τ	5×10^{-6}
Discrete speeds	c_i	0, ± 1 , ± 3
Internal-energy parameter	η_0	1.75
Domain length, diaphragm position	L_x, x_0	20, 10
Final time	t_{end}	0.15
Slope limiter	generalized minmod	$\theta = 1.2$
Time integrator	SSP-RK3	$\Delta t = \min(\text{CFL} \Delta x / 3, \tau / 4)$, CFL = 0.025
Grids	N_x	3,125; 6,250; 12,500; 25,000; 50,000
Boundary condition	—	zero gradient (transmissive)

2 Test problem: the Sod shock tube

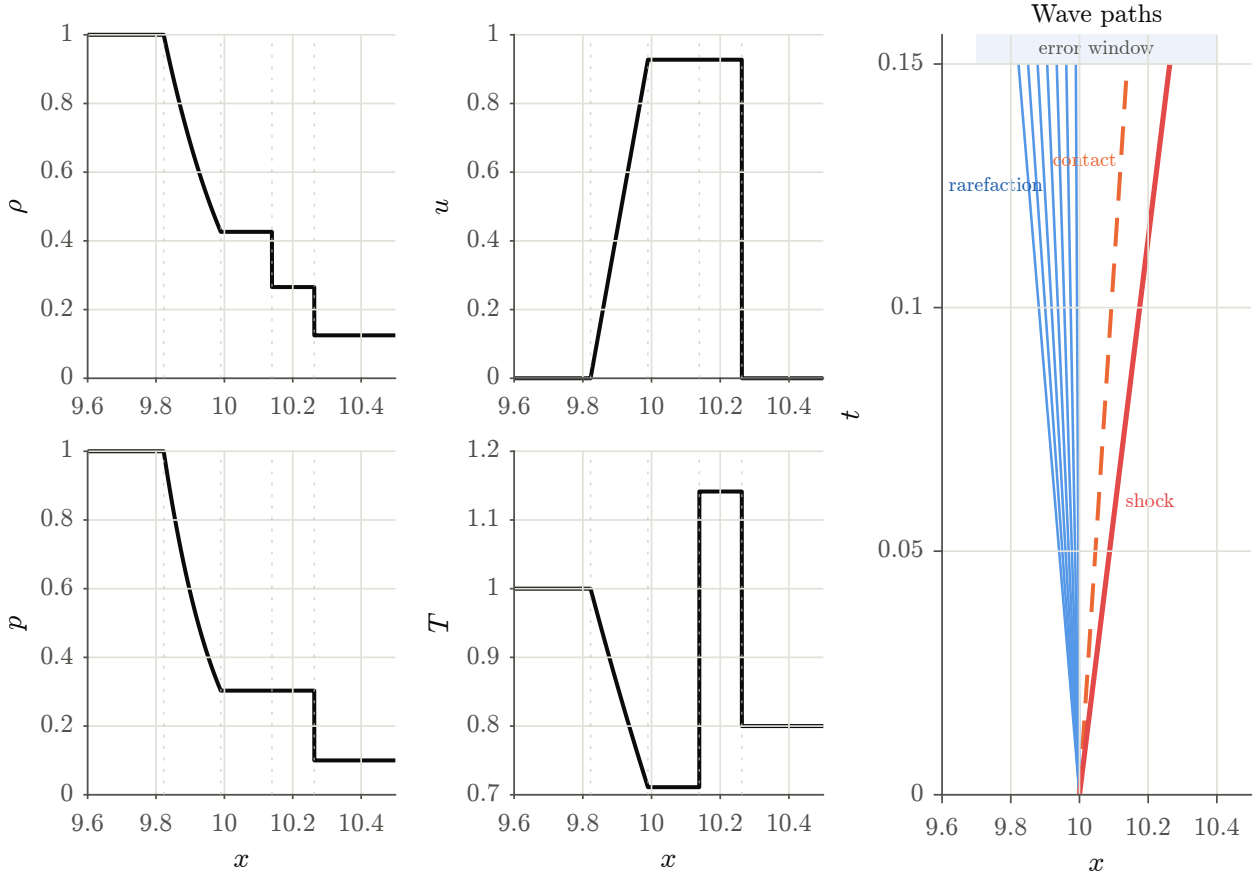
A tube of length 20 contains two gases separated by a diaphragm at $x_0 = 10$. The initial states are

$$(\rho, u, p) = (1, 0, 1) \text{ for } x < x_0, \quad (\rho, u, p) = (0.125, 0, 0.1) \text{ for } x > x_0,$$

where ρ is density, u velocity and p pressure (nondimensional, with $p = \rho T$). When the diaphragm is removed three waves form: a *rarefaction* travelling left, a *contact discontinuity* travelling right with the gas, and a *shock* travelling right ahead of it. The exact solution is obtained from the Riemann problem for the Euler equations [17]; its key values are listed in Table 2.

Table 2. Exact solution at $t = 0.15$.

Quantity	Value	Remark
Pressure and velocity between the waves	$p^* = 0.3031$, $u^* = 0.9275$	continuous across the contact
Density on either side of the contact	0.4263 / 0.2656	jump of 0.161
Temperature on either side of the contact	0.711 / 1.141	jump of 0.430
Shock speed (Mach number)	1.752 (1.66)	position $x_0 + 0.263$
Contact position	$x_0 + 0.139$	moves at u^*
Rarefaction extent	$x_0 - 0.177$ to $x_0 - 0.011$	smooth

**Figure 1.** Exact solution of the Sod problem at $t = 0.15$ (left) and the space–time paths of the three waves (right).**Discussion of Figure 1**

What is shown. The four panels on the left give density, velocity, pressure and temperature along the tube at the final time; dotted vertical lines mark the head and tail of the rarefaction, the contact and the shock. The right panel traces each wave in the x – t plane, starting from the diaphragm at $x = 10$, $t = 0$.

How to read it. In the wave diagram, the fan of thin blue lines is the rarefaction, the dashed orange line is the contact and the thick red line is the shock; steeper lines are slower waves. The shaded strip at the top marks the region $9.7 \leq x \leq 10.4$ used for the wave-region error

measurements.

What it means. The problem tests three distinct behaviours at once: a smooth expansion, a density/temperature jump with no pressure change (contact), and a jump in every variable (shock). Numerical methods typically handle these very differently, and the contact is usually the most difficult to keep sharp.

3 Physical model

3.1 Kinetic description

Instead of the fluid variables, the solver evolves a set of particle populations $f_i(x, t)$, each moving at a fixed speed c_i . Each population obeys the discrete BGK equation

$$\frac{\partial f_i}{\partial t} + c_i \frac{\partial f_i}{\partial x} = \frac{f_i^{\text{eq}} - f_i}{\tau}, \quad i = 1, \dots, 5. \quad (1)$$

The left-hand side describes free streaming of population i along the tube. The right-hand side is the Bhatnagar–Gross–Krook (BGK) collision model [1]: collisions relax each population toward a local equilibrium f_i^{eq} at a rate set by the relaxation time τ , which can be interpreted as the mean time between collisions. For small τ the populations remain close to equilibrium and their moments satisfy the Euler equations [2].

3.2 The D1V5 velocity set

The model uses one spatial dimension and five velocities (D1V5), following Kataoka and Tsutahara [9]:

$$c_i \in \{-3, -1, 0, +1, +3\}.$$

The stationary population is assigned an internal-energy parameter $\eta_0 = 1.75$. This represents the rotational and transverse energy of real molecules and allows the model to reproduce a gas with $\gamma = 1.4$.

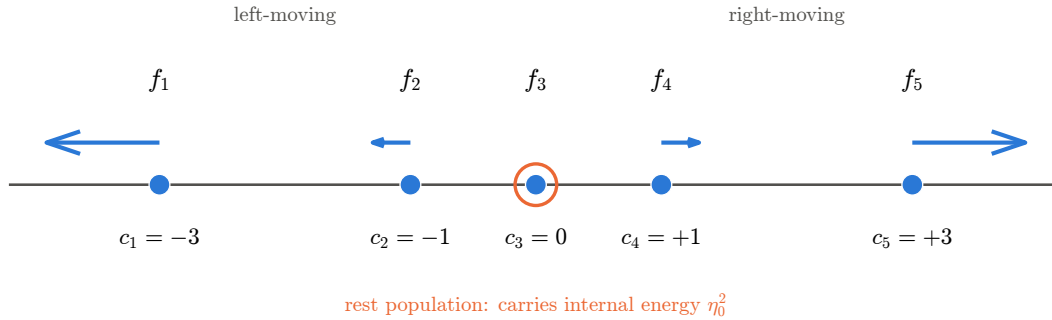


Figure 2. The five discrete velocities of the D1V5 model.

Discussion of Figure 2

What is shown. Each dot is one population f_i placed at its velocity c_i ; arrows indicate direction and relative speed. The circled centre population is stationary.

What it means. Two populations move left, two move right and one is at rest. The fastest speed, 3, sets the transport time-step limit (Section 4.4). Because the stationary population carries the internal energy, the model can represent a diatomic gas with only five populations.

3.3 Equilibrium distribution

The equilibrium populations $f_i^{\text{eq}}(\rho, u, T)$ are determined by requiring that they reproduce exactly five quantities: density, momentum, total energy, and the fluxes of momentum and energy that appear in the Euler equations. Five conditions on five unknowns give a unique solution, which is the explicit formula used by the solver. This equilibrium was verified numerically to reproduce all five quantities to machine precision ($\approx 10^{-15}$) over 2000 random states.

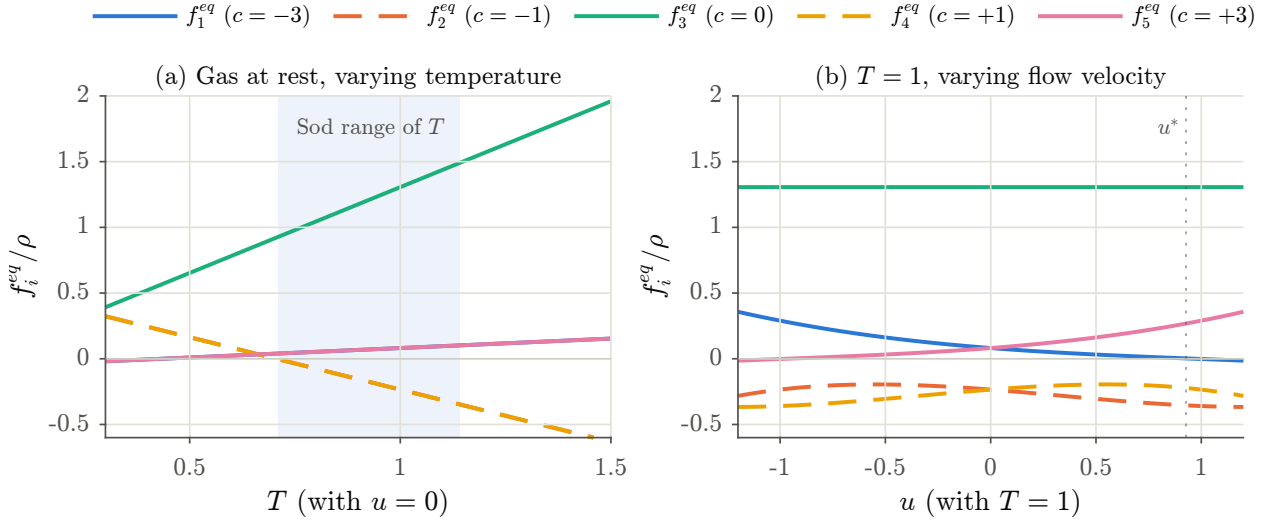


Figure 3. Equilibrium populations per unit density as functions of temperature (a) and flow velocity (b).

Discussion of Figure 3

What is shown. Panel (a) shows the five equilibrium values for a gas at rest as the temperature varies; the shaded band marks the temperatures that occur in the Sod solution. Panel (b) holds $T = 1$ and varies the flow velocity.

How to read it. In (a) the left- and right-moving pairs coincide because the gas is at rest. In (b) a positive velocity increases the right-moving populations and decreases the left-moving ones, which is how momentum is represented.

What it means. The slow populations ($c = \pm 1$) are *negative* over most of the Sod temperature range. This is permissible because the populations are moment carriers rather than physical particle counts, but it means positivity cannot be enforced on the individual f_i ; the solver instead checks that the recovered density and temperature remain positive. Panel (b) also shows that near the post-shock velocity u^* the fast population moving against the flow approaches zero.

3.4 Recovery of the macroscopic variables

The fluid variables in each cell are weighted sums of the five populations:

$$\rho = \sum_i f_i, \quad \rho u = \sum_i c_i f_i, \quad \rho (bT + u^2) = \sum_i (c_i^2 + \eta_i^2) f_i, \quad p = \rho T, \quad (2)$$

with $b = 2/(\gamma - 1) = 5$. Temperature is obtained by subtracting the kinetic-energy contribution u^2 from the total energy. Because it is the difference of two larger quantities, temperature is the variable most sensitive to small errors in individual populations, particularly the fast ones. This is why numerical overshoots appear first in the temperature profile (Section 6.7).

Collisions conserve mass, momentum and energy exactly, since the equilibrium is constructed from the same totals as the current populations.

3.5 Effect of the relaxation time

The exact Sod solution assumes an inviscid gas ($\tau = 0$), whereas the model uses a small but finite τ . A first-order (Chapman–Enskog) analysis of the D1V5 model shows that finite τ has two practical consequences:

- The physical shock thickness is approximately $7\tau \approx 4 \times 10^{-5}$, well below the finest cell size (4×10^{-4}). The computed shock width is therefore set by the grid, not by τ .
- The contact diffuses with time, like heat conduction, and reaches a physical width of about $1.2\text{--}1.5 \times 10^{-3}$ at $t = 0.15$ (about three cells on the finest grid). Grid refinement beyond this scale cannot sharpen the contact further.

The same analysis shows that the diffusion rate at a contact depends on the local flow speed and would become negative (unstable) if the flow speed exceeded the slow particle speed of 1. The D1V5 velocity set is therefore suited to flows with velocities below this value; the Sod problem ($u^* = 0.93$) lies just inside this limit.

4 Numerical method

4.1 Finite-volume discretization

The tube is divided into N_x cells of width $\Delta x = L_x/N_x$. Each cell stores the cell-averaged value of every population. Integrating Eq. (1) over a cell gives

$$\frac{df_{i,j}}{dt} = -\frac{F_{i,j+1/2} - F_{i,j-1/2}}{\Delta x} + \frac{f_{i,j}^{\text{eq}} - f_{i,j}}{\tau}, \quad (3)$$

where $F_{i,j\pm 1/2}$ is the flux of population i through the right and left faces of cell j . Because the flux leaving one cell enters its neighbour, the scheme is conservative by construction; the results confirm conservation of mass and energy to $\approx 10^{-12}$.

4.2 Upwind fluxes and MUSCL reconstruction

Each population moves in a single direction, so the flux at a face is taken from the upstream side (upwinding): populations with $c_i > 0$ use the value reconstructed from the cell on the left of the

face, those with $c_i < 0$ use the value from the right, and the stationary population has no flux. For linear advection this upwind choice is the exact solution of the local Riemann problem.

Using the cell average itself at the face gives a first-order scheme that is robust but strongly diffusive. MUSCL reconstruction [20] instead represents the data in each cell by a straight line of slope σ_j and evaluates that line at the faces:

$$f_{j+1/2}^- = f_j + \frac{1}{2}\sigma_j, \quad f_{j+1/2}^+ = f_{j+1} - \frac{1}{2}\sigma_{j+1}. \quad (4)$$

With an accurate slope, the leading diffusive error of the first-order scheme cancels and the method becomes second-order accurate in smooth regions.

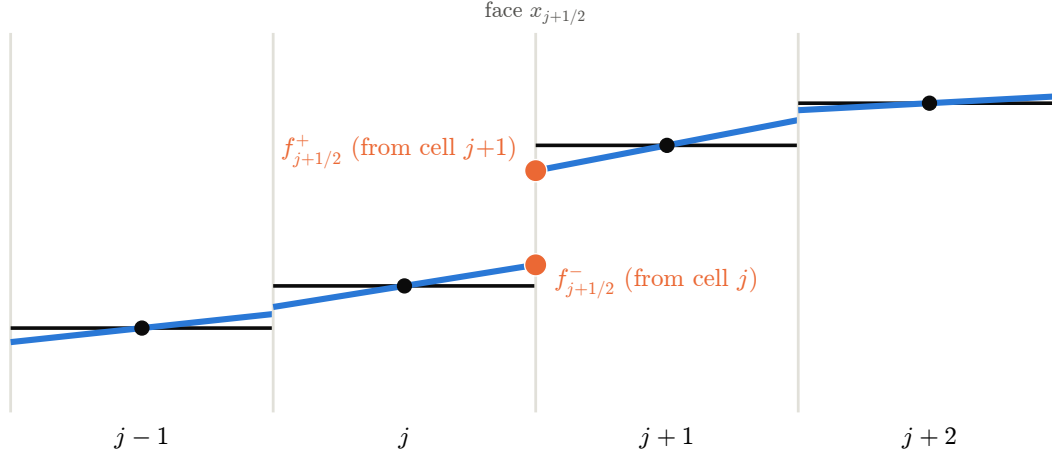


Figure 4. MUSCL reconstruction for one population on four cells.

Discussion of Figure 4

What is shown. Black horizontal segments are the stored cell averages; blue lines are the limited linear reconstructions within each cell; the two orange points are the two candidate values at the face between cells j and $j+1$.

How to read it. A right-moving population takes the lower orange value $f_{j+1/2}^-$ (from cell j); a left-moving population takes the upper value $f_{j+1/2}^+$ (from cell $j+1$). The gap between the two values is largest where the data change rapidly.

What it means. Compared with a first-order method, which would use the flat black values, the sloped reconstruction places a more accurate value at each face and greatly reduces numerical smearing. The slope must be limited near jumps to avoid creating new maxima or minima, which is the role of the limiter described next.

4.3 Generalized-minmod slope limiter

Let $d_L = f_j - f_{j-1}$ and $d_R = f_{j+1} - f_j$. The slope is

$$\sigma_j = \text{minmod}\left(\theta d_L, \frac{1}{2}(d_L + d_R), \theta d_R\right), \quad 1 \leq \theta \leq 2, \quad (5)$$

where minmod returns the argument of smallest magnitude if all arguments share the same sign, and zero otherwise [10]. The parameter θ controls how aggressive the reconstruction is: $\theta = 1$ reproduces the minmod limiter (most dissipative) and $\theta = 2$ the monotonized-central (MC) limiter [19]

(sharpest). The solver uses $\theta = 1.2$. For $1 \leq \theta \leq 2$ the scheme is total-variation diminishing (TVD) for each population's transport, meaning that it cannot create new oscillations [7, 16].

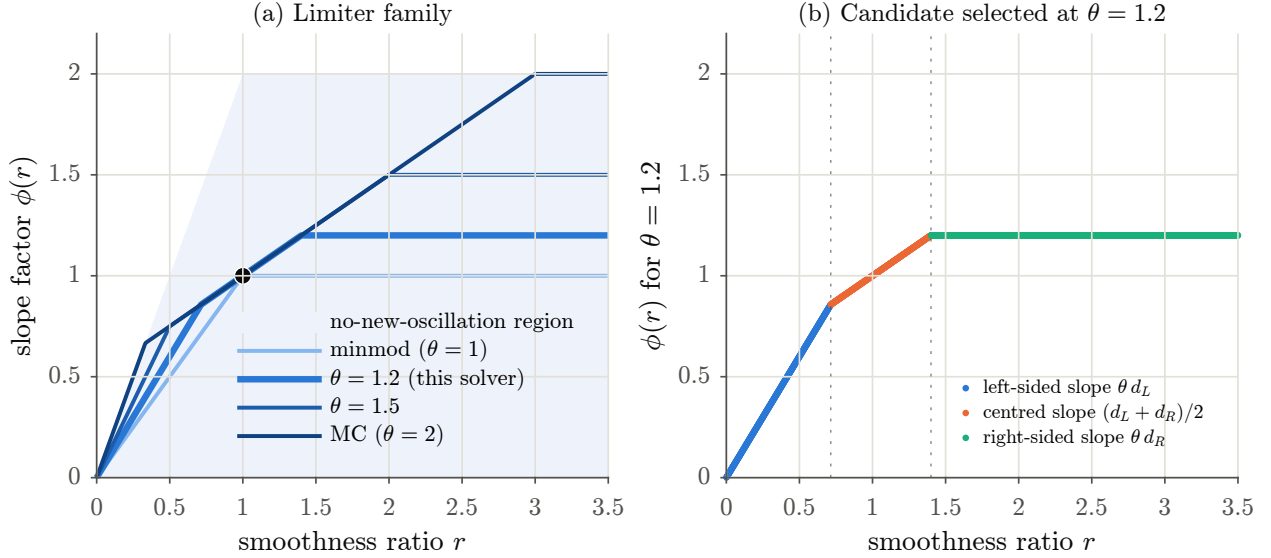


Figure 5. Generalized-minmod limiter functions (a) and the candidate slope selected at $\theta = 1.2$ (b).

Discussion of Figure 5

What is shown. The horizontal axis is the smoothness ratio $r = d_L/d_R$, which equals 1 for perfectly linear data and is negative at a local maximum or minimum. The vertical axis is the fraction $\phi(r) = \sigma/d_R$ of the right-hand difference used as the slope. The shaded area is the region in which a limiter is guaranteed not to create new oscillations.

How to read it. Higher curves allow steeper slopes (sharper results). All curves pass through $\phi(1) = 1$, so linear data are reproduced exactly. Panel (b) colours the $\theta = 1.2$ curve by which of the three candidate slopes in Eq. (5) is chosen.

What it means. $\theta = 1.2$ lies only slightly above minmod: it permits slopes up to 20% steeper than a one-sided difference and uses the centred slope for moderately smooth data ($0.71 < r < 1.4$). This yields a modest gain in sharpness over minmod while remaining far from the aggressive MC limiter, which is the source of the temperature overshoot discussed in Section 6.7.

4.4 Time integration and time-step selection

The semi-discrete system Eq. (3) is advanced with the three-stage, third-order strong-stability-preserving Runge–Kutta (SSP-RK3) scheme of Shu and Osher [3, 14]:

$$\begin{aligned} f^{(1)} &= f^n + \Delta t \mathcal{L}(f^n), \\ f^{(2)} &= \frac{3}{4}f^n + \frac{1}{4}[f^{(1)} + \Delta t \mathcal{L}(f^{(1)})], \\ f^{n+1} &= \frac{1}{3}f^n + \frac{2}{3}[f^{(2)} + \Delta t \mathcal{L}(f^{(2)})], \end{aligned}$$

where $\mathcal{L}$ denotes the right-hand side of Eq. (3). Each stage is a weighted average of forward-Euler steps, so the non-oscillatory property of the spatial scheme is preserved.

The time step is the smaller of a transport limit and a collision limit:

$$\Delta t = \min\left(\text{CFL} \frac{\Delta x}{\max |c_i|}, \frac{\tau}{4}\right). \quad (6)$$

With $\tau = 5 \times 10^{-6}$ the collision limit gives $\Delta t = 1.25 \times 10^{-6}$, which is smaller than the transport limit on every grid. All runs therefore use the same Δt and 120,000 time steps, and the effective transport Courant number is only 6×10^{-4} to 9×10^{-3} . This restriction is the dominant factor in the computational cost of the method (Section 6.4).

4.5 Robustness safeguards

After each Runge–Kutta stage the solver checks that density and temperature are positive and finite. A failed stage is repeated with half the time step and, if necessary, with a more dissipative limiter. No retries or limiter fallbacks occurred in any run reported here, so all results are those of the unmodified scheme.

5 Accuracy assessment

At every cell the computed value q_j is compared with the exact value q_j^{ex} :

$$L_1 = \frac{1}{N_x} \sum_j |q_j - q_j^{\text{ex}}|, \quad L_2 = \left[\frac{1}{N_x} \sum_j (q_j - q_j^{\text{ex}})^2 \right]^{1/2}. \quad (7)$$

L_1 is the mean absolute error; L_2 is the root-mean-square error, which weights large local errors more heavily. If the error behaves as $E \approx C \Delta x^p$, the *observed order* p between two grids is

$$p = \frac{\ln(E_{\text{coarse}}/E_{\text{fine}})}{\ln(\Delta x_{\text{coarse}}/\Delta x_{\text{fine}})}, \quad (8)$$

and a least-squares fit of $\ln E$ against $\ln \Delta x$ gives the order over all grids. Halving the cell size halves the error for $p = 1$ (first order) and quarters it for $p = 2$ (second order).

What order should be expected on this problem?

The formal order of a method (1 for first-order upwind, 2 for MUSCL) applies only to smooth solutions. A discontinuity is always smeared over a few cells, so the error there does not shrink as quickly. For methods of this type the expected orders on the Sod problem are [4, 6, 11]:

Method	Smooth solution		With shock and contact	
	L_1	L_2	L_1	L_2
First order	1	1	0.5 (contact) to 1 (shock)	0.25 to 0.5
Second order (MUSCL)	2	2	0.67 (contact) to 1 (shock)	0.33 to 0.5

Asymptotically, an L_1 order near 1 and an L_2 order near 0.5 is the best that any shock-capturing method, of any formal order, can achieve on this problem; on finite grids the measured values can differ somewhat from these limits.

6 Results

6.1 Solution on the finest grid

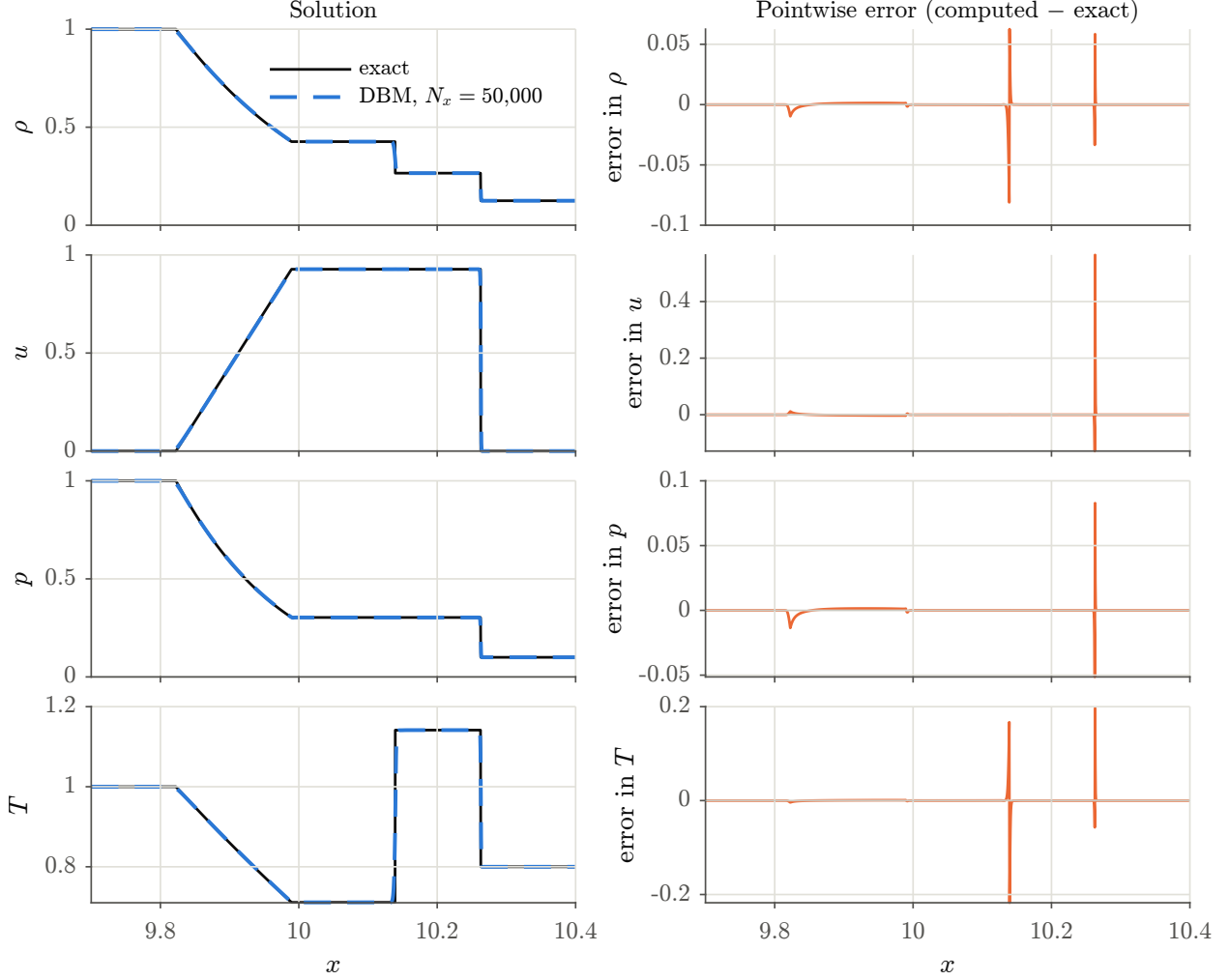


Figure 6. Computed solution on 50,000 cells compared with the exact solution (left), and the pointwise error (right).

Discussion of Figure 6

What is shown. The left column overlays the computed density, velocity, pressure and temperature (dashed blue) on the exact solution (black) in the region containing the waves. The right column plots the difference between them at each cell.

How to read it. Where the curves on the left coincide, the error on the right is zero. Spikes in the error column mark locations where the computed profile is slightly displaced or rounded relative to the exact jump.

What it means. The computed solution matches the exact one closely in all four variables and contains no spurious oscillations. The error is confined to a few cells at the contact (density and temperature) and at the shock (all variables), with small deviations at the corners of the

rarefaction. The peak pointwise error does not vanish under refinement because a jump spread over a few cells always leaves an $O(1)$ error in the cells that straddle it; the average errors (L_1 , L_2) nonetheless decrease, as shown below.

6.2 Effect of grid refinement

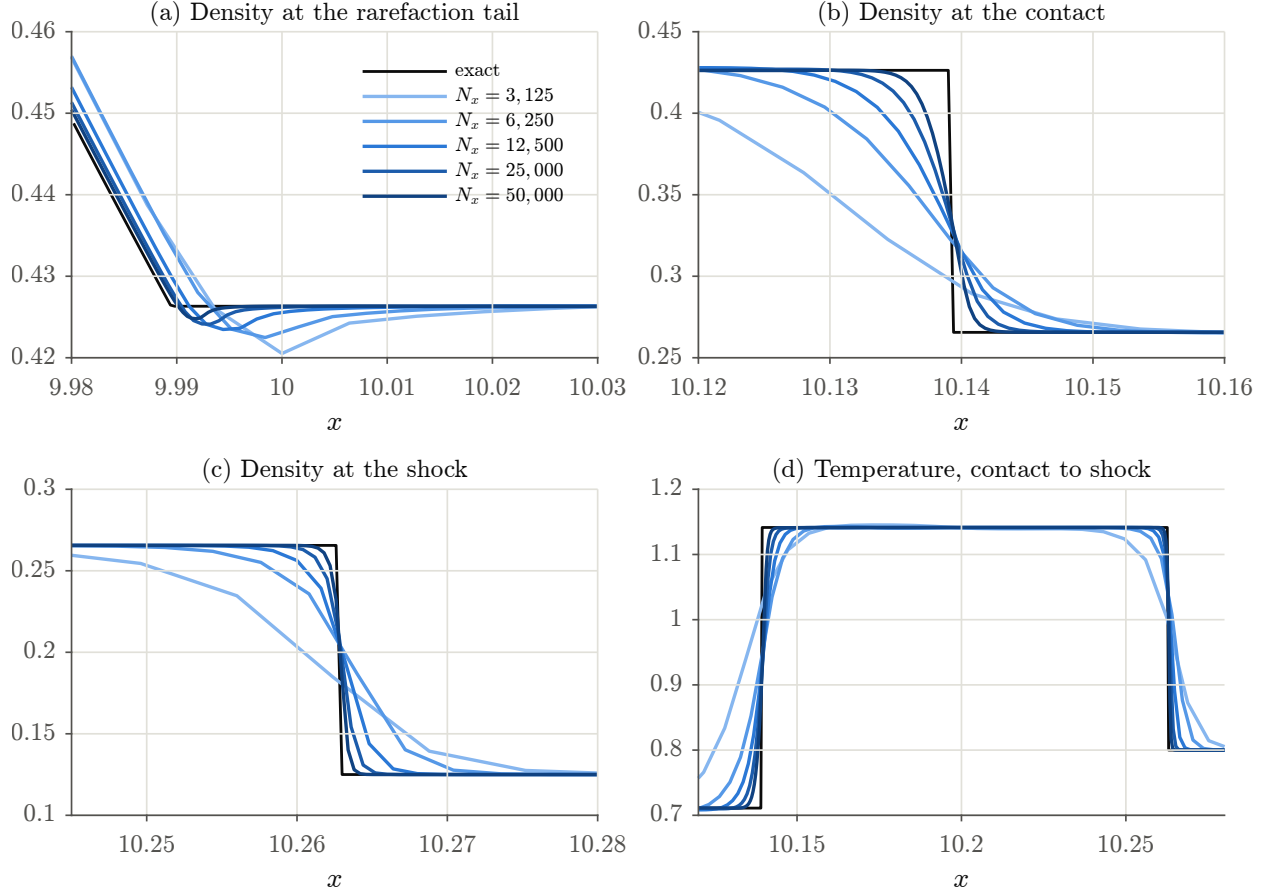


Figure 7. Close-up views of each wave on the five grids. Darker curves correspond to finer grids.

Discussion of Figure 7

What is shown. Magnified views of (a) the tail of the rarefaction, (b) the contact, (c) the shock and (d) the temperature between the contact and the shock, for all five grids.

How to read it. The black line is the exact solution; the lightest blue curve is the coarsest grid (3,125 cells) and the darkest is the finest (50,000 cells). A sharper step and a curve closer to black indicate higher accuracy.

What it means. Every refinement sharpens every wave. The shock (c) occupies roughly three cells on every grid, so its physical width halves with each refinement. The contact (b) narrows more slowly and spreads over more cells as the grid is refined, which is characteristic of contact discontinuities. A small temperature overshoot visible on the coarsest grids in (d) disappears by 12,500 cells. A slight undershoot below the tail of the rarefaction (a) also diminishes with

refinement.

6.3 Convergence and order of accuracy

Table 3. Global errors of the DBM solver ($\theta = 1.2$) and observed orders from a least-squares fit over all five grids.

N_x	L_1 error				L_2 error			
	ρ	u	p	T	ρ	u	p	T
3,125	2.46e-4	4.14e-4	2.14e-4	3.58e-4	3.16e-3	7.79e-3	3.08e-3	5.92e-3
6,250	1.30e-4	2.69e-4	1.09e-4	2.08e-4	1.81e-3	8.06e-3	1.64e-3	4.65e-3
12,500	7.23e-5	1.43e-4	5.75e-5	1.17e-4	1.39e-3	6.13e-3	1.14e-3	3.54e-3
25,000	4.07e-5	6.29e-5	3.11e-5	6.36e-5	1.00e-3	2.71e-3	7.20e-4	2.44e-3
50,000	2.51e-5	3.97e-5	1.88e-5	3.98e-5	7.77e-4	2.75e-3	5.15e-4	2.02e-3
Observed order	0.83	0.89	0.88	0.81	0.49	0.46	0.63	0.40

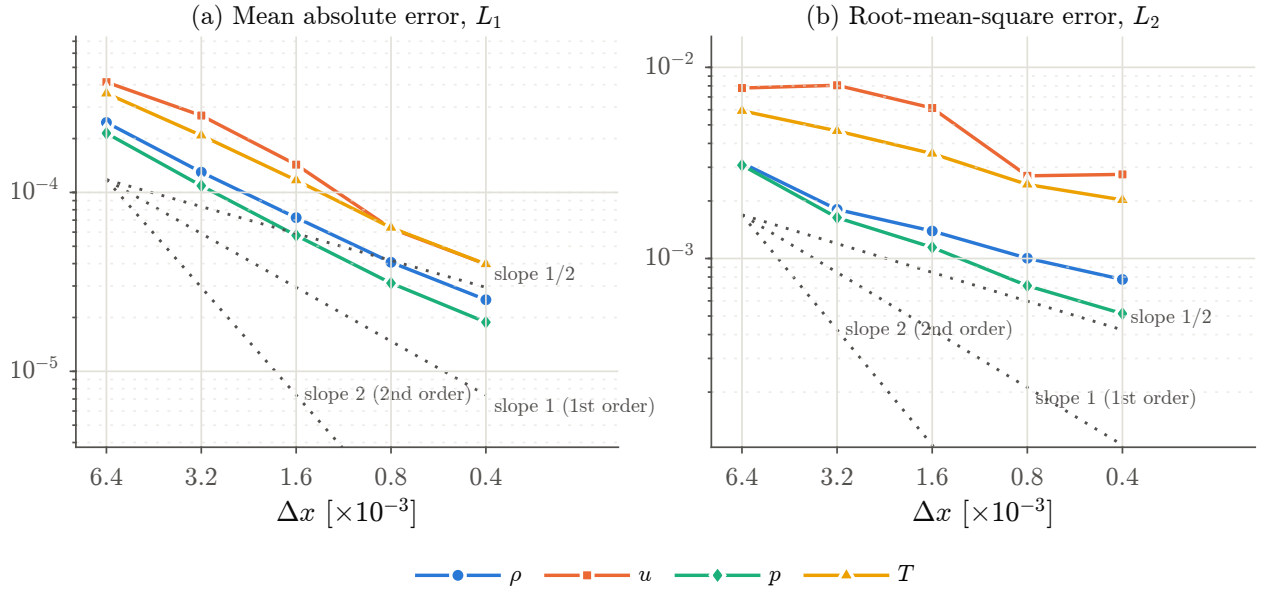


Figure 8. Convergence of the L_1 (a) and L_2 (b) errors of the DBM solver. Dotted lines show reference slopes of 2 (second order), 1 (first order) and $1/2$.

Discussion of Figure 8

What is shown. The error in each variable plotted against cell size on logarithmic axes, with the finest grid on the right. On such a plot a method of order p appears as a straight line of slope p .

How to read it. Compare the slope of each data line with the dotted reference lines. A line parallel to “slope 2” would indicate second-order convergence; parallel to “slope 1”, first order.

What it means. All four L_1 curves fall with a slope just below 1, and the L_2 curves with slopes between about 0.4 and 0.6, between the “slope $1/2$ ” and “slope 1” references. Both sets are

close to the best possible rates for a solution containing a shock and a contact (Section 5). The velocity curve in (b) levels off between the two finest grids; Section 6.5 shows that this is a measurement effect.

Does the L_2 error indicate a first-order or a second-order solution?

The measured L_2 orders are **0.40 to 0.63**, and the L_1 orders are **0.81 to 0.89**. Taken at face value, both are *below first order*; neither norm shows second-order convergence. This does not mean the method is less accurate than a first-order scheme. The reconstruction is formally second order, but on a problem with discontinuities every shock-capturing scheme is asymptotically limited to an L_1 order of about 1 and an L_2 order of about 1/2, and the contact lowers these further. The measured values should therefore be judged against other methods on the same problem rather than against the smooth-flow values of 1 and 2. Section 6.4 does this directly: on the same grids a first-order version of the solver converges with L_1 order 0.64, while the present solver (0.86) matches a conventional second-order Euler solver (0.92). The solver therefore behaves as a *second-order* method would on this problem: clearly better than first order, but, like every shock-capturing scheme, well below the formal rate of 2.

6.4 Comparison with other methods: accuracy and computational cost

To place the solver in context, the same Sod problem was solved on the same grids with five alternative methods:

- the **same DBM** with three other reconstructions: first-order upwind (no slope), MUSCL with minmod, and MUSCL with MC; and
- a **conventional finite-volume Euler solver**, which evolves density, momentum and energy directly, using the HLLC approximate Riemann solver [17, 18] with either first-order reconstruction or the same generalized-minmod MUSCL reconstruction ($\theta = 1.2$) and SSP-RK3 time stepping at a standard CFL number of 0.5.

All computations used the same computer, one processor core and MATLAB R2024b. The Euler times are the best of three measured runs. The DBM times are the measured cost per cell per time step (4.05×10^{-7} s for MUSCL and 2.74×10^{-7} s for first order, obtained from isolated runs on the two coarsest grids) multiplied by the number of cells and time steps; this cost was verified to be the same on both grids to within 0.4%.

Table 4. Accuracy and cost of the six methods. Orders are least-squares fits over the grids 3,125–25,000 (common to all methods); errors and cost are for 25,000 cells.

Method	Observed order (ρ)		Error at 25,000 cells		Cost at 25,000 cells	
	L_1	L_2	$L_1(\rho)$	$L_2(\rho)$	time steps	time [s]
DBM, first-order upwind	0.64	0.43	2.2×10^{-4}	2.5×10^{-3}	120,001	820
DBM, MUSCL minmod	0.80	0.49	5.6×10^{-5}	1.2×10^{-3}	120,001	1,215
DBM, MUSCL $\theta = 1.2$	0.86	0.53	4.1×10^{-5}	1.0×10^{-3}	120,001	1,215
DBM, MUSCL MC	0.87	0.55	3.5×10^{-5}	9.4×10^{-4}	120,001	1,215
Euler, first-order HLLC	0.64	0.41	1.8×10^{-4}	2.2×10^{-3}	818	4.0
Euler, MUSCL-HLLC $\theta = 1.2$	0.92	0.49	3.1×10^{-5}	9.5×10^{-4}	821	6.0

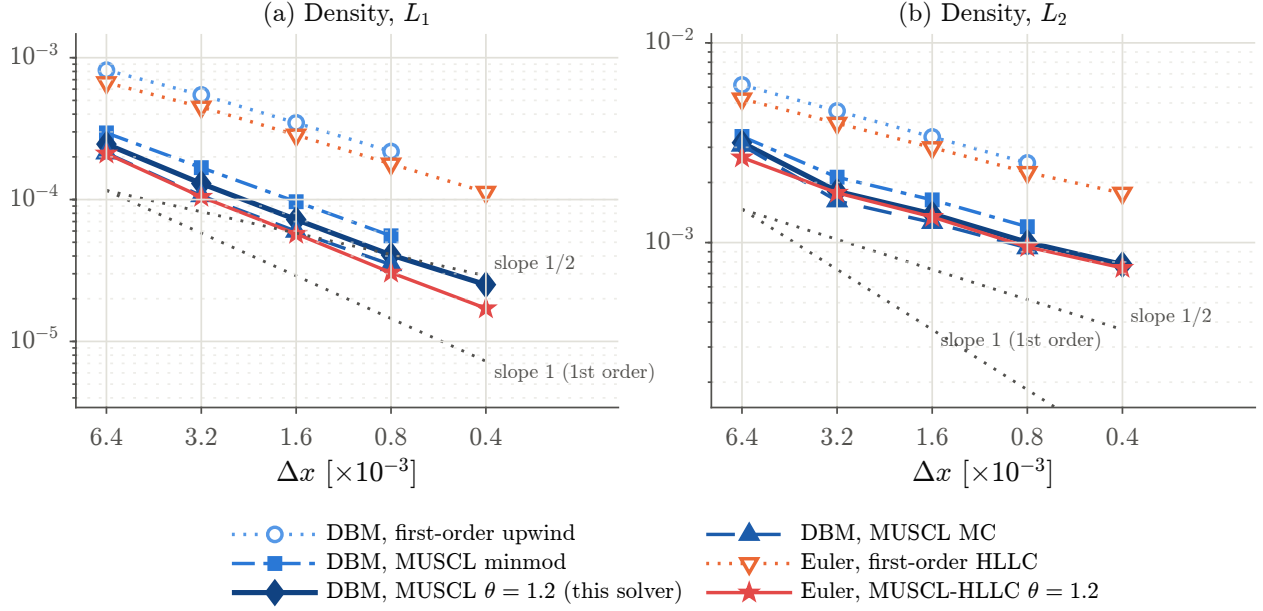


Figure 9. Density error versus cell size for the six methods: L_1 (a) and L_2 (b).

Discussion of Figure 9

What is shown. The density error of every method on every grid. Blue curves are DBM variants (the dark solid curve is the solver studied here); red and orange curves are the conventional Euler solver. Hollow markers denote first-order methods.

How to read it. Lower curves are more accurate at a given grid; steeper curves improve faster with refinement. The dotted lines give reference slopes of 1 and 1/2.

What it means. The methods separate into two groups. The two first-order methods (hollow markers) have errors three to seven times larger, with the gap widening under refinement, and converge more slowly (L_1 order 0.64). All second-order methods cluster together with L_1 orders of 0.80–0.92, whether they are kinetic (DBM) or conventional (Euler). At equal resolution the density error of the present DBM is about one third larger than that of the conventional MUSCL-HLLC solver. The order of accuracy is therefore determined by the reconstruction, not by whether the gas is modelled kinetically or with the Euler equations.

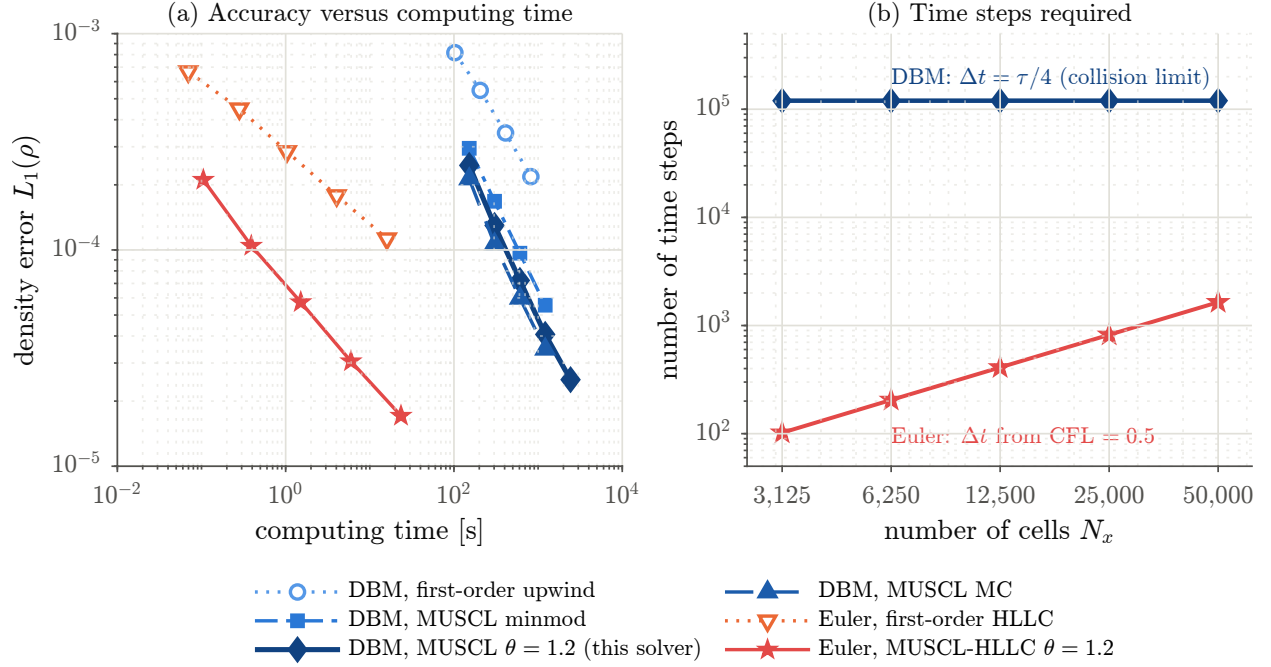


Figure 10. (a) Density error versus computing time for the six methods. (b) Number of time steps required by the DBM and the Euler solver on each grid.

Discussion of Figure 10

What is shown. Panel (a) is a work–precision diagram: each point is one grid, placed by its computing time (horizontal) and its density error (vertical). Panel (b) shows how many time steps each solver needed on each grid.

How to read it. In (a), the most efficient method is the one whose curve lies furthest to the lower left, meaning the lowest error for a given time. A horizontal line through the plot compares the time each method needs to reach the same error.

What it means. The conventional Euler solvers are two to three orders of magnitude faster than the DBM for the same accuracy. Panel (b) explains why. The cost per cell per time step is similar for both solvers (4.1×10^{-7} s for the DBM versus about 2.9×10^{-7} s for MUSCL-HLLC), but the DBM needs 120,001 steps on every grid because its time step is fixed by the collision limit $\Delta t = \tau/4$, whereas the Euler solver’s step is set by the wave speed and needs only 102 steps on the coarsest grid and 1,643 on the finest. The gap therefore narrows with refinement, from about 780 times at 6,250 cells to about 100 times at 50,000 cells.

Quantitative summary.

- **Second-order versus first-order DBM.** At the same grid, MUSCL with $\theta = 1.2$ reduces the density error by a factor of 5.4 for only 1.5 times the cost per step. Reaching the same accuracy with first-order upwinding would require roughly 14 times as many cells and about 9 times the computing time.
- **Choice of limiter.** MC is about 15% more accurate on average than $\theta = 1.2$ at identical cost, but produces the temperature overshoot described in Section 6.7; minmod has about

35% larger error.

- **DBM versus conventional Euler solver.** On 50,000 cells the DBM reaches $L_1(\rho) = 2.5 \times 10^{-5}$ in an estimated 2,430 s. The MUSCL-HLLC solver reaches the same error on about 32,000 cells in roughly 10 s, about 250 times less. At equal resolution its density error is 25% smaller (at 25,000 cells).
- **Source of the cost.** The difference is entirely due to the number of time steps. If the DBM time step were set by the transport CFL condition ($\text{CFL} = 0.5$) instead of by $\tau/4$, it would need about 2,250 steps on 50,000 cells instead of 120,001, a reduction by a factor of about 50. This requires treating the collision term implicitly [13] or using a scheme that couples transport and collision [5].

Context from the literature. Table 5 places these results alongside other widely used shock-capturing approaches. An important reference point is the study of Greenough and Rider [4], who compared a fifth-order WENO scheme [8] with a second-order piecewise-linear Godunov (MUSCL-type) scheme. For problems with discontinuities, including the Sod problem, both schemes converged at approximately first order; on Sod the second-order scheme had about half the error of WENO and was about six times cheaper at equal resolution. Comparative studies of many schemes on standard Euler test problems [12] similarly show that no method avoids the accuracy limit imposed by discontinuities. A second-order limited reconstruction is therefore a sound choice for shock-dominated flows, and the present DBM uses one. Its efficiency disadvantage arises from the explicit treatment of collisions, not from the spatial scheme.

Table 5. Qualitative comparison of shock-capturing approaches for problems such as Sod.

Method	Formal order	Accuracy and cost on shock problems
First-order upwind / Godunov	1	Robust but strongly diffusive; errors 3–7 times larger than second-order methods at equal resolution (Table 4).
MUSCL / TVD with limiter [16, 20]	2	Standard choice; convergence limited by discontinuities to about first order in L_1 ; cheapest route to a given accuracy on shock problems [4].
WENO5 [8]	5	Far more accurate for smooth flow; on Sod, about first-order convergence with larger error and about six times higher cost than a second-order Godunov scheme at equal resolution [4].
Conventional Euler MUSCL-HLLC (this study)	2	$L_1(\rho)$ order 0.92; about 250 times cheaper than the DBM for equal accuracy on this problem.
FVDBM, explicit collision (this study)	2	Accuracy close to the conventional second-order solver; time step limited to $\tau/4$, giving 120,001 steps on every grid. Provides kinetic information not available from the Euler equations.
Kinetic schemes with implicit or coupled collision [5, 13]	2	Remove the $\Delta t \propto \tau$ restriction; the expected route to making the DBM competitive in cost.

6.5 Velocity error and the position of the shock within a cell

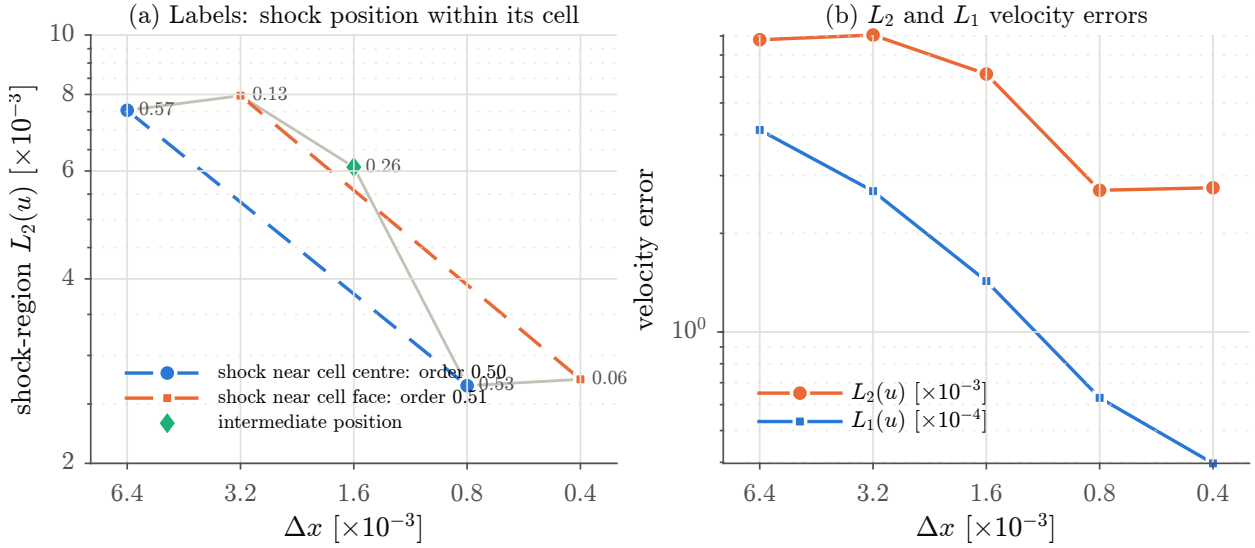


Figure 11. (a) Velocity error in the cells surrounding the shock, labelled with the position of the exact shock within its cell. (b) Global L_2 and L_1 velocity errors.

Discussion of Figure 11

What is shown. Panel (a) shows the L_2 velocity error contributed by the eight cells around the shock. Each point is labelled with the fractional position of the exact shock inside its cell (0 = at a cell face, 0.5 = at the cell centre). Panel (b) compares the global L_2 and L_1 velocity errors.

How to read it. Grids on which the shock falls at a similar position within a cell are joined by the same coloured line; the legend reports the observed order along each line.

What it means. Between 93% and 99% of the velocity L_2 error comes from the cells at the shock, where the velocity drops from 0.93 to 0. The size of that error depends on whether the exact shock lies near a cell face or near a cell centre, and this position differs from grid to grid. When grids with comparable shock positions are compared, the error decreases at the expected order of about 0.5. The apparent stall of $L_2(u)$ between 25,000 and 50,000 cells is therefore an artefact of the error measurement, not a limitation of the solver. The L_1 velocity error, which is less sensitive to a few cells, decreases steadily with order 0.89.

6.6 Contact and shock thickness; influence of the relaxation time

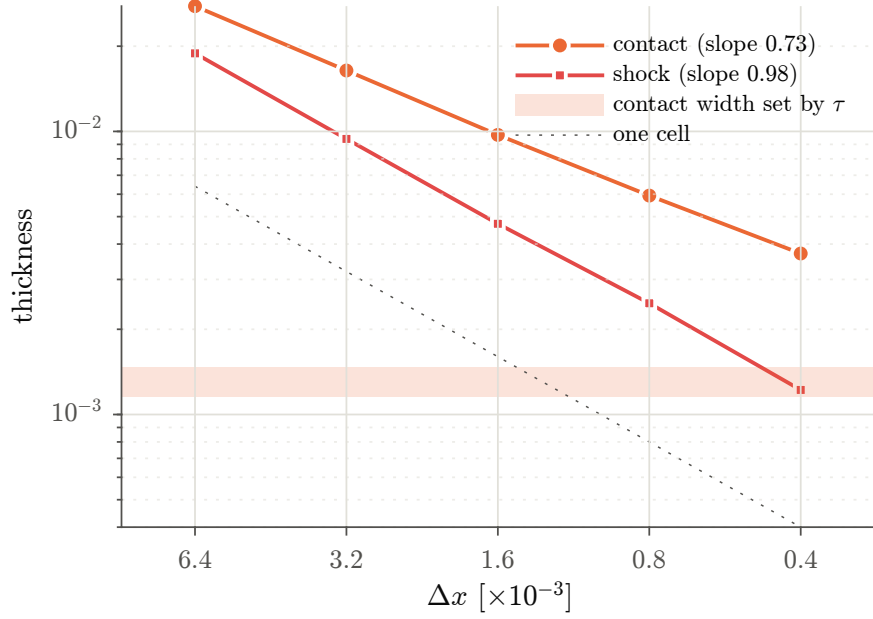


Figure 12. Measured thickness of the contact and the shock versus cell size. The shaded band is the contact width set by the relaxation time τ .

Discussion of Figure 12

What is shown. The thickness of each discontinuity, defined as its jump divided by the steepest computed gradient, for each grid. The dotted line marks a thickness of one cell; the shaded band marks the physical contact width produced by $\tau = 5 \times 10^{-6}$.

How to read it. A slope of 1 means the feature always occupies the same number of cells. A feature that reached the shaded band would be limited by the physics of the model rather than by the grid.

What it means. The shock has slope 0.98 and spans about three cells on every grid, as expected for a captured shock. The contact has slope 0.73, close to the theoretical $2/3$ for a second-order scheme, and grows from 4 to 9 cells across the grid sequence. On the finest grid the contact is still about three times wider than its physical (τ -set) width, so the present results are limited by grid resolution rather than by τ . Extrapolating the measured trend, the two widths would become equal at about 2×10^5 cells.

The influence of τ is confirmed by comparison with an earlier study at a ten times larger relaxation time ($\tau = 5 \times 10^{-5}$; Table 6). With the larger τ the contact is naturally about three times wider, and the density error stopped decreasing at approximately 1.13×10^{-3} beyond 40,000 cells. With the present value it continues to decrease. The earlier runs also used a larger time step, so this comparison changes two parameters at once.

Table 6. Global L_2 density error for two relaxation times.

N_x	$\tau = 5 \times 10^{-5}$ (earlier study)	$\tau = 5 \times 10^{-6}$ (present)	Ratio
3,125	3.24×10^{-3}	3.16×10^{-3}	0.97
6,250	1.96×10^{-3}	1.81×10^{-3}	0.92
12,500	1.59×10^{-3}	1.39×10^{-3}	0.87
25,000	1.26×10^{-3}	1.00×10^{-3}	0.80
40,000	1.13×10^{-3}	—	—
50,000	1.13×10^{-3}	0.78×10^{-3}	0.69

6.7 Limiter comparison

Five limiters developed during the project were compared on 6,250 cells with all other settings identical (Table 7 and Figure 13). The two “hybrid” limiters switched between MC and minmod, either separately for each population or using a shared indicator based on density, pressure and temperature.

Table 7. Limiter comparison on 6,250 cells. Lower values are better in every column.

Limiter	$L_1(\rho)$	$L_1(u)$	$L_1(p)$	$L_1(T)$	T overshoot
Minmod	1.68e-4	3.36e-4	1.42e-4	2.65e-4	4.5e-4
MC	1.06e-4	2.07e-4	8.86e-5	1.68e-4	6.7e-3
Population hybrid	1.34e-4	2.89e-4	1.14e-4	2.16e-4	2.6e-3
Macroscopic hybrid	1.64e-4	3.27e-4	1.35e-4	2.66e-4	6.2e-4
Generalized minmod, $\theta = 1.2$	1.30e-4	2.69e-4	1.09e-4	2.08e-4	1.6e-3

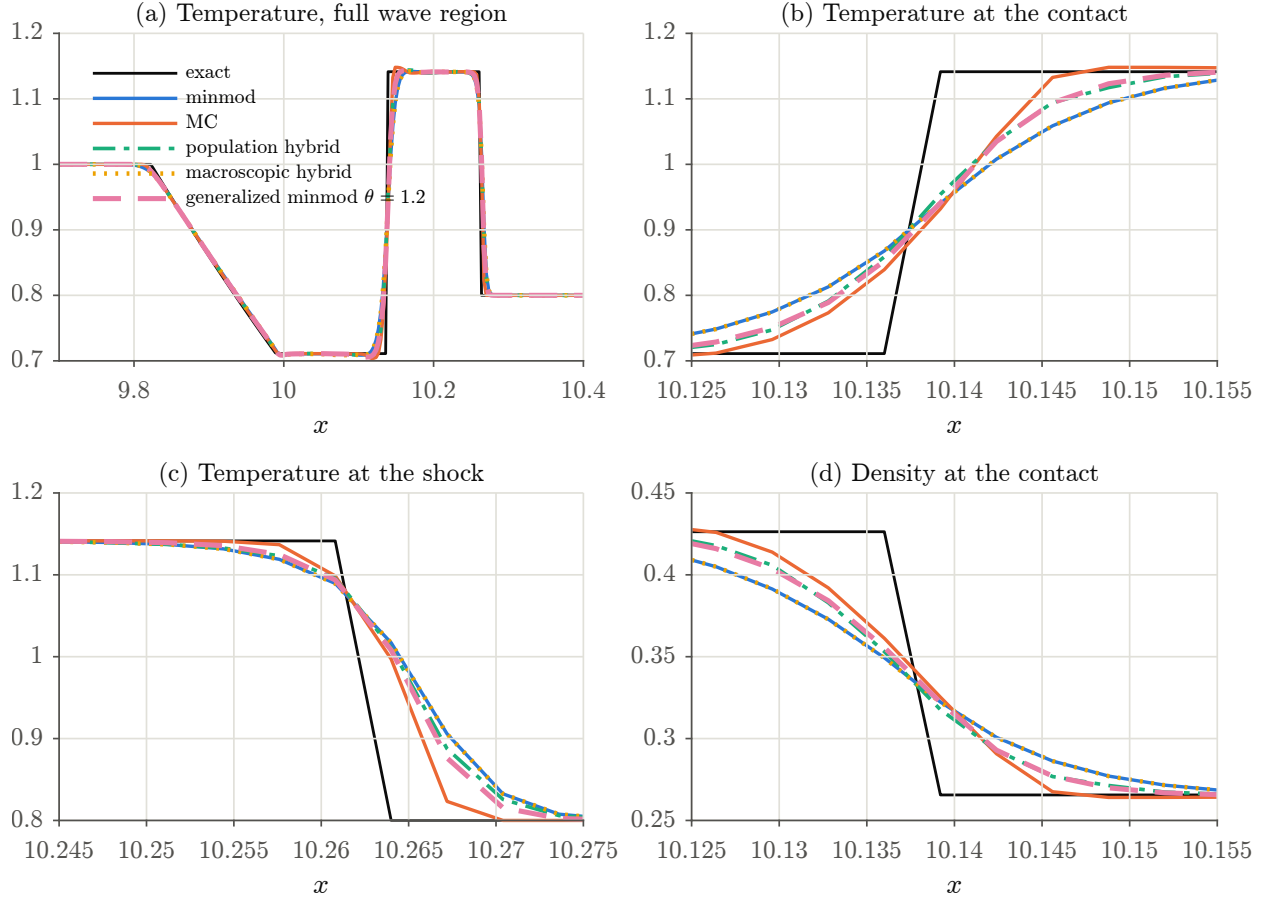


Figure 13. Temperature and density profiles near the contact and shock for the five limiters on 6,250 cells.

Discussion of Figure 13

What is shown. Panel (a) shows temperature over the whole wave region; panels (b)–(d) magnify the temperature at the contact and the shock, and the density at the contact.

How to read it. A curve that rises more steeply is sharper (less diffusive). A curve that rises above the black exact value just behind the contact is overshooting.

What it means. MC (orange) is the sharpest but exceeds the exact post-contact temperature, an overshoot of 6.7×10^{-3} . Minmod (blue) and the macroscopic hybrid (nearly identical to minmod) do not overshoot but are the most diffused. Generalized minmod with $\theta = 1.2$ (pink dashed) is visibly sharper than minmod while keeping the overshoot small: it lowers all four L_1 errors by 20–23% relative to minmod and reduces the MC overshoot by 77%.

6.8 Selection of the limiter parameter θ

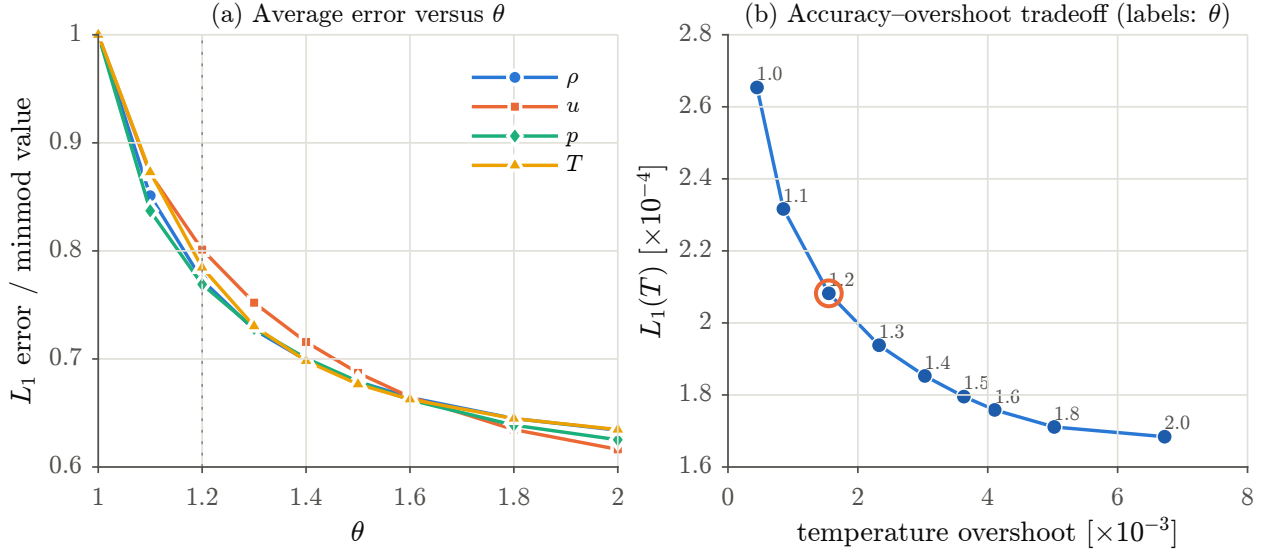


Figure 14. Effect of θ on average error (a) and the tradeoff between temperature error and temperature overshoot (b), on 6,250 cells.

Discussion of Figure 14

What is shown. Nine values of θ between 1 (minmod) and 2 (MC). Panel (a) shows each variable's L_1 error relative to its minmod value; panel (b) plots the temperature error against the temperature overshoot, with each point labelled by θ .

How to read it. In (a), lower is better. In (b), the ideal point would be the lower-left corner (small error and small overshoot); the circled point is $\theta = 1.2$.

What it means. Increasing θ always reduces the average error and always increases the overshoot; no value is best in both respects. The curve in (b) is steep at small θ , so the first increments of θ buy large accuracy gains at little overshoot cost. $\theta = 1.2$ obtains 59% of the error reduction available between minmod and MC while incurring 18% of the overshoot increase. On finer grids the overshoot at $\theta = 1.2$ disappears entirely (Figure 7d), making it a sound working choice.

7 Conclusions and recommendations

1. The D1V5 finite-volume DBM solves the Sod problem accurately and robustly: no oscillations, no stability interventions, and conservation of mass and energy to 10^{-12} on all grids.
2. The observed orders (L_1 : 0.81–0.89; L_2 : 0.40–0.63) are below first order in the strict sense, but they match a conventional second-order Euler solver on the same grids and clearly exceed first-order methods (L_1 order 0.64); this is the expected behaviour of a second-order scheme on a solution with a shock and a contact.
3. The contact discontinuity limits convergence; the shock remains about three cells wide on every grid. At $\tau = 5 \times 10^{-6}$ the results remain grid-limited up to 50,000 cells.

4. At equal resolution the DBM and a conventional second-order Euler solver achieve similar accuracy, but the DBM is 100 to 800 times more expensive because its explicit collision term restricts the time step to $\tau/4$; the cost per time step is comparable. The method's advantage lies in its kinetic (non-equilibrium) information rather than in efficiency for purely inviscid problems.
5. Generalized minmod with $\theta = 1.2$ provides a balanced compromise between numerical diffusion and temperature overshoot.

Recommendations.

- Remove the collision limit on the time step, either by treating the collision term implicitly with an IMEX Runge–Kutta scheme [13] or by adopting a scheme that couples transport and collision, such as the discrete unified gas-kinetic scheme [5]. The time step would then be set by the transport CFL condition rather than by τ .
- Perform a single time-step-halving test on the finest grid to confirm that temporal error is negligible.
- Verify formal second-order accuracy on a smooth test problem, such as a small-amplitude acoustic pulse, which the Sod problem cannot demonstrate.
- Report velocity convergence using the L_1 norm, or with grid sequences that sample different shock positions, to avoid the measurement effect of Section 6.5.
- Before extending the method to two dimensions, select a velocity set whose slowest speed exceeds the expected flow velocities.

Nomenclature

c_i	discrete particle velocity	p	pressure
f_i	population (distribution) i	T	temperature
f_i^{eq}	equilibrium population	u	flow velocity
τ	relaxation time	ρ	density
η_0	internal-energy parameter	γ	ratio of specific heats
$\Delta x, N_x$	cell size, number of cells	Δt	time step
θ	limiter parameter	CFL	Courant number
L_1, L_2	mean-absolute and RMS errors	p	also: observed order of accuracy

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